Engineering Requirements Specification

Generic Consumer Home Vacuum Robot (Early Model) — Reference Model GEN-RVC-100

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1. Introduction

1.1 Purpose

This Engineering Requirements Specification defines the functional, performance, mechanical, electrical, navigation, acoustic, environmental, and software requirements for the Generic Consumer Home Vacuum Robot, an early-model robotic floor cleaner (hereafter referred to as the "robot"). The document establishes the technical baseline used by hardware, firmware, mechanical, electrical, navigation, acoustic, and validation engineering teams during the product development life cycle.

1.2 Scope

The robot is a consumer-grade autonomous floor cleaning appliance intended for residential use on hard floors and low-pile carpet. The product consists of the robot body with integrated rechargeable battery, a charging dock, a virtual wall accessory (an infrared barrier emitter), an infrared remote control, and a starter set of replacement filters and brushes. The specification covers the design, performance characteristics, verification requirements, and validation methodologies for the robot and its accessories as a system.

The robot is explicitly an early-generation design. It uses a random-bounce navigation pattern supplemented by wall-following and spot-cleaning behaviors. It does not include simultaneous localization and mapping, optical or laser ranging, on-board cameras, Wi-Fi connectivity, or a companion mobile application. Localization is short-horizon and relies on contact bump sensors, infrared cliff sensors, wheel encoders, and an infrared dock-homing beacon.

1.3 Intended Audience

The intended audience includes program managers, systems engineers, mechanical engineers, electrical engineers, firmware engineers, navigation engineers, acoustic engineers, test engineers, quality engineers, and reliability engineers participating in the product development life cycle.

1.4 Reference Documents

The following reference documents are cited for context. Each reference is a publicly available specification, standard, or technical resource.

Identifier	Title	Source
IEC 60312-1	Vacuum cleaners for household use — Methods of measuring performance	https://webstore.iec.ch/publication/22013
IEC 62885 series	Surface cleaning appliances	https://webstore.iec.ch/publication/26152
IEC 60335-2-2	Household appliance safety — Vacuum cleaners and water-suction appliances	https://webstore.iec.ch/publication/63645
IEC 60335-2-108	Household appliance safety — Robotic cleaners (battery-operated)	https://webstore.iec.ch/publication/65789
IEC 60704-2-1	Acoustic noise test code for vacuum cleaners	https://webstore.iec.ch/publication/26352
IEC 60068	Environmental Testing	https://webstore.iec.ch/publication/505
IEC 60529	Degrees of Protection Provided by Enclosures (IP Code)	https://webstore.iec.ch/publication/2452

Identifier	Title	Source
IEC 62133-2	Safety requirements for portable sealed secondary Li-Ion cells	https://webstore.iec.ch/publication/32662
IEC 61951-2	Sealed NiMH cells and batteries for portable use	https://webstore.iec.ch/publication/24232
UN 38.3	Transport of dangerous goods — Tests for lithium batteries	https://unece.org/transport/dangerous-goods
ANSI/AHAM VC-1	Method of measuring vacuum cleaner performance	https://www.aham.org
EN 1822 series	High-efficiency air filters (EPA, HEPA, ULPA)	https://www.cencenelec.eu
ISO 5725	Accuracy of measurement methods and results	https://www.iso.org/standard/69419.html

1.5 Definitions, Acronyms, and Abbreviations

Term	Definition
AW	Air Watt — unit of suction performance defined by IEC 60312.
BLDC	Brushless DC (Direct Current) motor.
BMS	Battery Management System.
CFM	Cubic Feet per Minute.
EMC	Electromagnetic Compatibility.
ESD	Electrostatic Discharge.
FW	Firmware.
HEPA	High Efficiency Particulate Air filter (per EN 1822, classes H13–H14).
IMU	Inertial Measurement Unit.
IP (Code)	Ingress Protection rating per IEC 60529.
IR	Infrared.
LCD	Liquid Crystal Display.
LED	Light Emitting Diode.
Li-Ion	Lithium Ion rechargeable battery chemistry.
MCU	Microcontroller Unit.
MTBF	Mean Time Between Failures.
NiMH	Nickel Metal Hydride rechargeable battery chemistry.
NTC	Negative Temperature Coefficient thermistor.
OCP	Over-Current Protection.
OTP	Over-Temperature Protection.
OVP	Over-Voltage Protection.

Term	Definition
PCB	Printed Circuit Board.
PMU	Power Management Unit.
PWM	Pulse Width Modulation.
RH	Relative Humidity.
RPM	Revolutions Per Minute.
SOC	State of Charge (of battery).
SOH	State of Health (of battery).
TOF	Time of Flight (sensing).
UVP	Under-Voltage Protection.
UI	User Interface.

2. System Overview

2.1 Product Concept

The robot is a battery-powered autonomous floor cleaner with a disc-shaped body roughly 340 mm in diameter and 90 mm in height. Two differentially-driven drive wheels and a passive front caster provide mobility. A single horizontally mounted main brush, a single side brush, and a suction motor draw debris into a 0.4 L dust bin. A 14.4 V NiMH battery pack provides power for approximately 60 minutes of continuous operation on hard floor. The robot navigates the room using a random-bounce pattern interleaved with wall-following along detected obstacles and a localized spot-clean spiral. When battery state of charge falls below a defined threshold, the robot homes to a charging dock using an infrared beacon and resumes charging until full. The product also includes a remote control and a virtual wall barrier accessory.

The design is explicitly an early-generation robotic floor cleaner. It does not perform simultaneous localization and mapping. It does not include laser ranging, time-of-flight ranging, on-board cameras, an inertial measurement unit, Wi-Fi, Bluetooth, or a companion mobile application. All user interaction is performed through on-body buttons, an infrared remote, on-body LED indicators, and an audible beeper.

2.2 System Block Diagram

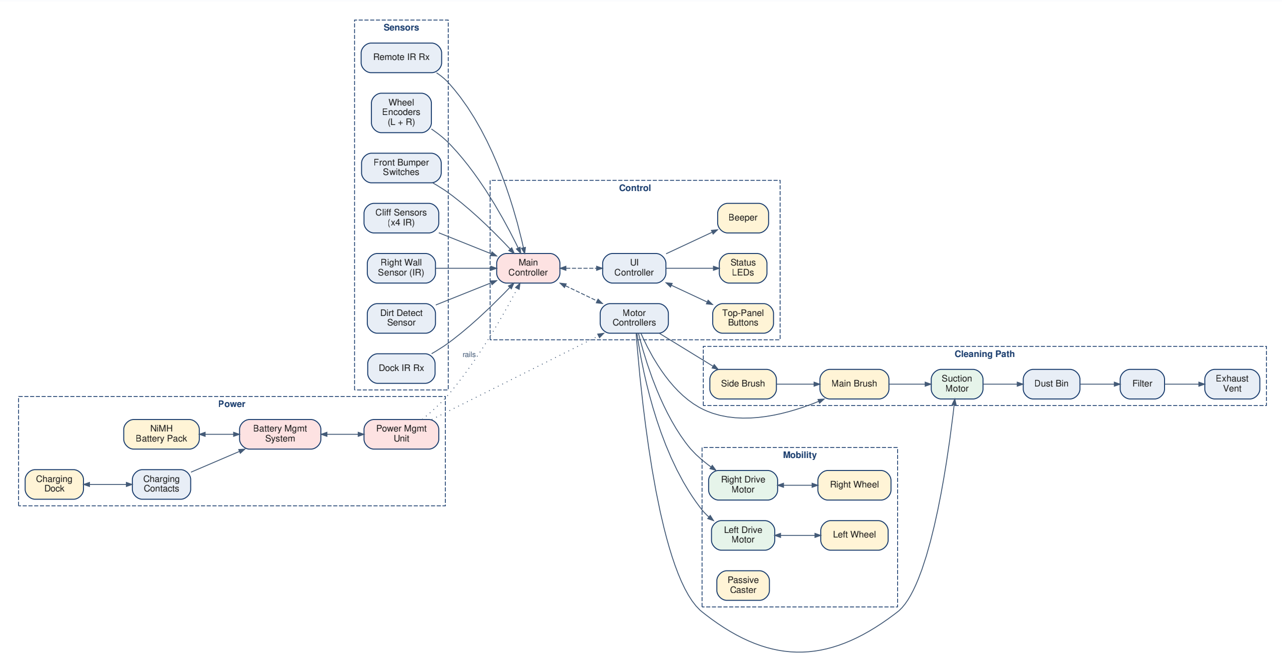

Figure 2-1. System Block Diagram.

2.3 Airflow Path Diagram

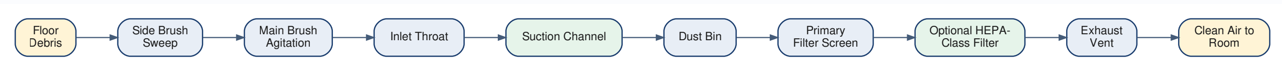

Figure 2-2. Airflow Path Diagram.

2.4 Sensor Placement Diagram

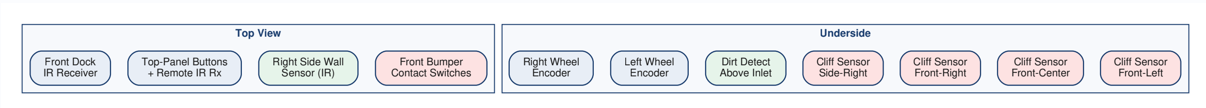

Figure 2-3. Sensor Placement Layout.

2.5 Operating Modes

Mode	Description
Off	Robot off the dock; main controller in deep sleep; BMS in monitor-only state.
Standby	Robot on the dock; battery fully charged or trickle-charging; UI dimmed.
Charging	Robot on the dock; battery actively charging.
Clean	Robot performing autonomous cleaning using random-bounce, wall-follow, and spot behaviors.
Spot	Robot performing a localized 1 m diameter spiral spot-clean for ~3 min.
Dock	Robot has detected low battery or completed cleaning; homing to dock using IR beacon.
Pause	User has paused the current cleaning cycle; robot stationary; UI active.

Mode	Description
Fault	Wheel stall, brush stall, cliff hazard hold-off, or battery fault active; UI shows fault code.

3. Functional Requirements

Functional requirements describe what the robot shall do. Each requirement carries a unique identifier of the form FR-XXX, a priority designation, and a verification method. Priority levels are Critical (the product cannot ship without this function), High (the product cannot meet its market positioning without this function), and Medium (the function meaningfully improves the product but the product is still viable without it). Verification methods are Test (T), Analysis (A), Inspection (I), and Demonstration (D).

3.1 Cleaning Performance Functional Requirements

ID	Requirement	Priority	Verification
FR-001	The robot shall provide suction power ≥ 15 AW at nominal battery state of charge.	Critical	T
FR-002	The robot shall include a counter-rotating main brush and a separately driven side brush.	Critical	I
FR-003	The robot shall capture loose debris from hard floor at a minimum pickup rate of 90 % by mass per IEC 62885-3 standardized test.	Critical	T
FR-004	The robot shall capture loose debris from low-pile carpet at a minimum pickup rate of 70 % by mass per IEC 62885-3 standardized test.	High	T
FR-005	The robot shall include a removable dust bin with capacity ≥ 0.4 L.	Critical	I
FR-006	The robot dust bin shall be removable and emptied by the user without contact with the filter.	High	D
FR-007	The robot shall enter a localized spot-clean spiral mode when the Spot button is pressed.	High	D

3.2 Navigation and Mobility Functional Requirements

ID	Requirement	Priority	Verification
FR-101	The robot shall navigate autonomously using a random-bounce pattern combined with wall-follow and spot-clean behaviors.	Critical	D
FR-102	The robot shall detect and stop at floor drop-offs of ≥ 50 mm using downward-facing cliff sensors.	Critical	T
FR-103	The robot shall detect contact with obstacles via a front bumper assembly and change direction within 200 ms of bumper depression.	Critical	T
FR-104	The robot shall follow a continuous wall on its right side at an offset of 30 mm to 50 mm using an infrared wall sensor.	High	T
FR-105	The robot shall traverse hard floor surfaces at a nominal forward speed of 0.3 m/s.	High	T
FR-106	The robot shall traverse transitions of ≤ 15 mm between flooring surfaces without becoming stuck.	High	T

ID	Requirement	Priority	Verification
FR-107	The robot shall return to its charging dock using an infrared beacon when the battery state of charge falls below 15 %.	Critical	T
FR-108	The robot shall respect an infrared virtual wall barrier and shall not cross the barrier line during cleaning.	High	T
FR-109	The robot shall automatically stop and indicate a fault if any drive wheel stalls for ≥ 2 s.	Critical	T
FR-110	The robot shall automatically stop and indicate a fault if either brush stalls for ≥ 2 s.	Critical	T

3.3 Power and Runtime Functional Requirements

ID	Requirement	Priority	Verification
FR-201	The robot shall provide a continuous cleaning runtime ≥ 60 min on hard floor at nominal SOC.	Critical	T
FR-202	The robot shall provide a continuous cleaning runtime ≥ 45 min on low-pile carpet at nominal SOC.	High	T
FR-203	The robot shall charge from 0 % to 100 % SOC in ≤ 3.5 h on the charging dock.	Critical	T
FR-204	The robot shall display a 3-level battery state-of-charge indicator.	High	I
FR-205	The robot shall provide a low-battery indication at SOC ≤ 15 %.	High	T
FR-206	The robot shall transition autonomously to Dock mode when SOC ≤ 15 % during a cleaning cycle.	Critical	T

3.4 User Interface Functional Requirements

ID	Requirement	Priority	Verification
FR-301	The robot shall provide three top-panel buttons: Clean, Spot, and Dock.	High	I
FR-302	The robot shall accept commands from a paired infrared remote control.	High	D
FR-303	The robot shall provide a daily and weekly cleaning schedule with a single start time per day.	Medium	T
FR-304	The robot shall preserve schedule and last operating mode across battery removal and replacement, retaining settings for ≥ 30 days without battery.	Medium	T
FR-305	The robot shall provide LED indicators for power, mode, charge state, and fault status.	High	I
FR-306	The robot shall provide an audible beeper that issues distinct tones for start, dock complete, error, and battery low events.	High	D
FR-307	The robot shall encode fault conditions in a documented LED blink pattern and beeper tone pattern.	High	I

3.5 Filtration and Dust Handling Functional Requirements

ID	Requirement	Priority	Verification
FR-401	The robot shall include a primary filter screen at the dust bin outlet.	Critical	I
FR-402	The robot shall accept an optional HEPA-class filter element installable in place of the primary screen.	High	I
FR-403	The dust bin shall be transparent or have a translucent panel to allow visual assessment of fill level.	High	I
FR-404	The robot shall provide a dust-bin-full indication when the bin exceeds a defined fill threshold.	Medium	T
FR-405	The filter element shall be removable and washable by the user.	High	D

4. Specifications

The specifications table consolidates measurable performance parameters used throughout the design and verification process. Specifications are grouped by subsystem and each entry includes a target value, an acceptable tolerance, and a reference to the verification method that establishes compliance.

4.1 Cleaning Performance Specifications

Parameter	Target Value	Tolerance	Verification Method
Suction Power (Cleaning Mode)	15 AW	−2 AW	IEC 60312-1
Sealed Suction (Max Vacuum)	5 kPa	−0.5 kPa	Manometer at sealed inlet
Cleaning Width (Main Brush)	165 mm	± 3 mm	Caliper Measurement
Cleaning Width (with Side Brush Sweep)	250 mm	± 5 mm	Test Pattern Measurement
Main Brush Speed	1,000 rpm	± 50 rpm	Optical Tachometer
Side Brush Speed	90 rpm	± 10 rpm	Optical Tachometer
Suction Motor Speed	12,000 rpm	± 500 rpm	Optical Tachometer
Dust Bin Capacity	0.4 L	−0.02 L	Volumetric Fill
Hard Floor Pickup	≥ 90 %	–	IEC 62885-3
Low-Pile Carpet Pickup	≥ 70 %	–	IEC 62885-3
Edge Cleaning Reach	≥ 25 mm beyond chassis	–	Test Pattern Measurement

4.2 Mobility and Navigation Specifications

Parameter	Target Value	Tolerance	Verification Method
Nominal Forward Speed	0.3 m/s	± 0.05 m/s	Floor Test, Wheel Encoder Calibration
Drive Wheel Diameter	60 mm	± 0.5 mm	Caliper Measurement

Parameter	Target Value	Tolerance	Verification Method
Drive Wheel Travel (Suspension)	10 mm	± 1 mm	Suspension Travel Measurement
Climb Capability (Threshold)	15 mm	–	Threshold Test Bar
Cliff Detection Distance	≥ 50 mm drop	± 5 mm	Cliff Test Step
Cliff Sensor Response Time	≤ 100 ms	+ 20 ms	Cliff Test with Logic Analyzer
Bumper Activation Force	1.5 N	± 0.5 N	Force Gauge
Bumper Travel	6 mm	± 1 mm	Caliper Measurement
Wall-Follow Offset	30 mm to 50 mm	–	Wall-Follow Test Run
Dock IR Acquisition Range	≥ 4 m	–	Open-Floor Dock Test
Virtual Wall IR Range	2.5 m	± 0.5 m	Open-Floor Beam Test
Bumper Cycle Life	≥ 200,000 actuations	90 % statistical confidence	Life Cycle Test
Drive Wheel Encoder Resolution	16 ticks/revolution	Fixed	Encoder Specification

4.3 Mechanical Specifications

Parameter	Target Value	Tolerance	Verification Method
Body Mass (with Battery)	3.5 kg	± 0.1 kg	Weighing
Body Diameter	Ø 340 mm	± 2 mm	Caliper Measurement
Body Height	90 mm	± 1 mm	Caliper Measurement
Chassis Material	Glass-fiber reinforced polymer	Fixed	Inspection
Top Cover Material	Unreinforced polymer with rubberized buttons	Fixed	Inspection
Main Brush Bristle Length	25 mm	± 1 mm	Caliper Measurement
Side Brush Diameter	110 mm	± 2 mm	Caliper Measurement
Dust Bin Material	Transparent polymer	Fixed	Inspection
Carrying Handle Pull Strength	≥ 50 N	–	Pull Test
Trim Ring Snap-Fit Retention	≥ 30 N	–	Pull Test

4.4 Electrical Specifications

Parameter	Target Value	Tolerance	Verification Method
Battery Type	NiMH, 12-cell stick pack	Fixed	Inspection
Battery Nominal Voltage	14.4 V	± 0.5 V	Multimeter
Battery Capacity	2,200 mAh	−5 %	Discharge Test
Battery Energy	31.7 Wh	−2 Wh	Calculated
Battery Cycle Life	≥ 400 cycles to 80 % rated capacity	–	Cycle Life Test
Charging Dock Input	100–240 V AC, 50/60 Hz	–	Multimeter
Charging Dock Output	22 V DC, 1.25 A	± 0.5 V / ± 0.1 A	Power Analyzer
Charging Time (0 → 100 %)	3.5 h	+ 0.5 h	Charge Profile Log
Standby Current (On Dock)	< 10 mA	± 2 mA	Current Measurement
Standby Current (Off Dock)	< 200 µA	± 50 µA	Current Measurement
MCU Clock Frequency	16 MHz	± 0.5 MHz	Logic Analyzer
Motor Controller Switching Frequency	20 kHz PWM	± 2 kHz	Oscilloscope

4.5 Acoustic Specifications

Parameter	Target Value	Verification Method
Sound Power Level (Cleaning Mode)	≤ 68 dB(A)	IEC 60704-2-1
Sound Power Level (Docked, Charging)	≤ 30 dB(A)	IEC 60704-2-1
Beeper Sound Pressure Level at 1 m	65 to 75 dB(A)	Sound Level Meter

4.6 Environmental Specifications

Parameter	Target Value	Verification Method
Operating Temperature	5 °C to +40 °C	Climatic Chamber, per IEC 60068
Storage Temperature	−10 °C to +50 °C	Climatic Chamber, per IEC 60068
Operating Humidity	10 % to 85 % RH, non-condensing	Humidity Chamber, per IEC 60068
Mechanical Shock Survival	0.5 m drop onto hardwood, 3 orientations	Drop Test, per IEC 60068-2-31

Parameter	Target Value	Verification Method
Vibration Resistance	5 Hz to 100 Hz sweep, 0.5 g acceleration	Vibration Table, per IEC 60068-2-6
ESD Immunity	± 8 kV contact, ± 15 kV air	ESD Simulator
EMC	Compliant with regional emissions and immunity standards	Anechoic Chamber
Ingress Protection Rating	IP20	Per IEC 60529

5. Requirements

This section captures non-functional and constraint requirements that bound how the robot shall be designed, built, and qualified. Functional requirements are captured in section 3; specifications are captured in section 4. The requirements in this section govern reliability, safety, regulatory compliance, manufacturability, and serviceability.

5.1 Reliability Requirements

ID	Requirement	Priority	Verification
REQ-R-001	The robot shall achieve an MTBF ≥ 1,500 operating hours under typical consumer use.	Critical	A
REQ-R-002	The suction motor shall achieve ≥ 800 operating hours to first wear-out failure with 90 % confidence.	Critical	T
REQ-R-003	Each drive motor shall achieve ≥ 1,000 operating hours to first wear-out failure with 90 % confidence.	Critical	T
REQ-R-004	The bumper switch assembly shall survive ≥ 200,000 actuations.	Critical	T
REQ-R-005	The main brush shall survive ≥ 200 h of operation before bristle replacement.	High	T
REQ-R-006	The side brush shall survive ≥ 100 h of operation before bristle replacement.	High	T
REQ-R-007	The dust bin latch shall survive ≥ 5,000 open/close cycles.	High	T
REQ-R-008	The cliff sensor optical window shall not lose more than 10 % of its transmission after 200 h of exposure to typical floor debris and lint.	High	T
REQ-R-009	The charging contacts shall survive ≥ 5,000 docking cycles without exceeding contact resistance specification.	High	T
REQ-R-010	The wheel encoders shall not lose more than 1 % of their pulse count accuracy over the full life of the robot.	High	A

5.2 Safety Requirements

ID	Requirement	Priority	Verification
REQ-S-001	The robot shall not present a touchable surface temperature > 55 °C during any continuous operating mode.	Critical	T
REQ-S-002	The robot battery system shall include OVP, UVP, OCP, OTP, and short-circuit protection per IEC 62133-2 or IEC 61951-2 as applicable.	Critical	T
REQ-S-003	The robot shall stop all motors within 2 s of a detected cliff hazard.	Critical	T
REQ-S-004	The robot shall not present accessible voltages > 60 V DC.	Critical	T
REQ-S-005	The robot enclosure shall not present sharp edges or pinch points accessible to the user.	High	I
REQ-S-006	The robot brush enclosures shall be designed such that user contact with rotating brushes during operation is not possible.	Critical	A
REQ-S-007	The robot shall stop all motors within 1 s of being lifted off the floor surface.	Critical	T
REQ-S-008	The robot shall prevent operation when the dust bin is not seated.	High	T
REQ-S-009	The robot shall comply with IEC 60335-2-108 particular requirements for battery-operated robotic cleaners.	Critical	T

5.3 Regulatory and Compliance Requirements

ID	Requirement	Priority	Verification
REQ-C-001	The robot and its charging dock shall comply with regional EMC regulations for the target sales regions.	Critical	T
REQ-C-002	The robot shall comply with regional RoHS regulations.	Critical	A
REQ-C-003	The robot shall comply with regional product safety regulations for household appliances (e.g., IEC 60335-2-2, IEC 60335-2-108, UL 1017, or equivalent).	Critical	T
REQ-C-004	The robot battery shall pass UN 38.3 tests if Li-Ion cells are substituted in a future revision; for the base NiMH configuration, the battery shall meet IEC 61951-2 safety requirements.	Critical	T
REQ-C-005	The infrared remote, dock beacon, and virtual wall accessory shall comply with regional regulations for unlicensed infrared emitters.	Critical	T
REQ-C-006	The robot shall display all required regulatory marks in a permanent and legible location.	High	I
REQ-C-007	The charging dock shall comply with regional energy efficiency standby power regulations.	High	T

5.4 Manufacturing Requirements

ID	Requirement	Priority	Verification
REQ-M-001	The robot shall be assembled with a target assembly cycle time ≤ 18 min/unit at full production rate.	High	A
REQ-M-002	The robot bill of materials shall achieve a target unit material cost ≤ program-defined target.	High	A
REQ-M-003	The motor and motor controller assemblies shall be tested at the subassembly level before final assembly.	Critical	I
REQ-M-004	The battery pack shall be tested at the subassembly level for capacity, internal resistance, and cell balance before final assembly.	Critical	I
REQ-M-005	The completed robot shall pass an end-of-line functional test including drive, suction, navigation sample, and acoustic spot check.	Critical	T
REQ-M-006	The robot firmware version shall be programmable at end of line and updateable via a wired service port in the field.	High	D
REQ-M-007	The cliff sensor calibration shall be performed at end of line on a fixture that emulates the worst-case dark and light floor reflectance values.	Critical	T

5.5 Serviceability Requirements

ID	Requirement	Priority	Verification
REQ-V-001	The battery pack shall be replaceable by the user without tools in ≤ 60 s.	High	D
REQ-V-002	The dust bin shall be removable by the user without tools in ≤ 10 s.	Critical	D
REQ-V-003	The filter element shall be removable by the user without tools in ≤ 20 s.	High	D
REQ-V-004	The main brush and side brush shall be replaceable by the user with a single common service tool.	High	D
REQ-V-005	The drive wheel assembly shall be replaceable by a service technician in ≤ 30 min.	Medium	D
REQ-V-006	The robot shall log usage counters for total run time, mode usage distribution, fault events, drop-detection events, and stall events, retrievable through a service interface.	High	T

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6. Subsystem Requirements

6.1 Cleaning Subsystem

The cleaning subsystem comprises the side brush, the main brush, the inlet throat, the suction motor and impeller, the dust bin, the primary filter screen, the optional HEPA-class filter, and the exhaust vent. The subsystem governs how debris is swept, agitated, drawn into the bin, and retained.

ID	Requirement
SUB-C-001	The side brush shall sweep debris from beyond the chassis edge toward the inlet of the main brush.
SUB-C-002	The main brush shall agitate carpet fibers and lift debris into the inlet throat.
SUB-C-003	The main brush bristle pattern shall include a mix of stiff bristles for agitation and softer flicker bristles for debris transfer.
SUB-C-004	The inlet throat geometry shall maintain ≥ 4 kPa of pressure differential across the throat in cleaning mode.
SUB-C-005	The suction motor and impeller shall be balanced to ≤ 1 g·mm residual imbalance.
SUB-C-006	The dust bin shall include a tamper geometry that does not allow loose debris to fall back into the inlet during deceleration or backup.
SUB-C-007	The primary filter screen shall capture particles ≥ 50 µm at ≥ 95 % efficiency when clean.
SUB-C-008	The optional HEPA-class filter shall meet HEPA H13 per EN 1822 (≥ 99.95 % at MPPS).
SUB-C-009	The exhaust vent shall direct discharge air parallel to the floor surface and not directly at the dust bin opening.

6.2 Mobility Subsystem

The mobility subsystem comprises the two drive wheel assemblies (motor, gearbox, wheel, suspension, encoder), the passive front caster, the chassis, the suspension travel limits, and the wheel drop detection.

ID	Requirement
SUB-M-001	Each drive wheel assembly shall include a brushed DC motor, a reduction gearbox, a rubber-tired wheel, a spring-loaded suspension arm, and an incremental rotary encoder.
SUB-M-002	The wheel suspension shall provide ≥ 10 mm of vertical travel to accommodate threshold crossings.
SUB-M-003	Each drive wheel shall include a wheel-drop switch that asserts when the wheel is fully extended (robot lifted off floor).
SUB-M-004	The drive system shall be capable of stationary 360° in-place rotation with no lateral translation.
SUB-M-005	The drive motors shall include stall detection by current sensing on the motor controller bridge.
SUB-M-006	The passive caster shall include a debris guard that prevents string and hair from wrapping around the caster post.
SUB-M-007	The chassis shall be designed such that no projection extends below the lowest cliff sensor optical aperture.
SUB-M-008	The drive belts (if used in gearbox stages) shall be tensioned to manufacturer specification and retained against vibration loosening.

6.3 Sensing Subsystem

The sensing subsystem comprises the front bumper switches, the four downward infrared cliff sensors, the right-side infrared wall sensor, the dirt detection sensor, the dock infrared receiver, the remote infrared receiver, and the wheel encoders.

ID	Requirement
SUB-S-001	The front bumper shall include at least two independent contact switches arranged to provide left, center, and right collision detection.
SUB-S-002	The four cliff sensors shall be located at the front-left, front-center, front-right, and right-side positions on the chassis underside.
SUB-S-003	Each cliff sensor shall be a reflective infrared emitter-detector pair calibrated to the worst-case floor reflectance variation.
SUB-S-004	The right-side wall sensor shall be a single-axis infrared emitter-detector pair providing a continuous distance estimate within a defined range.
SUB-S-005	The dirt detection sensor shall be a piezoelectric impact sensor mounted above the inlet throat that triggers a localized spot-clean when impact rate exceeds a threshold.
SUB-S-006	The dock infrared receiver shall demodulate the dock beacon signal at the dock-defined carrier frequency and reject ambient infrared interference.
SUB-S-007	The remote infrared receiver shall demodulate remote command signals at a separate carrier frequency from the dock beacon.
SUB-S-008	The cliff sensor optical apertures shall include a lint-shedding geometry to minimize accumulation.

6.4 Electrical Subsystem

The electrical subsystem comprises the main PCB, the motor controller PCB, the PMU, the BMS, the battery pack and its contacts, the charging dock circuitry, the indicator LEDs, the beeper, and all internal wiring harnesses.

ID	Requirement
SUB-E-001	The PMU shall provide regulated power rails at 3.3 V and 5.0 V from the battery bus.
SUB-E-002	The PMU shall enter a sleep state with quiescent current < 200 µA within 60 s of cleaning end with the robot off the dock.
SUB-E-003	The MCU shall include a hardware watchdog timer that resets the system upon firmware hang.
SUB-E-004	The motor controllers shall provide closed-loop speed control on the drive motors using encoder feedback.
SUB-E-005	The motor controllers shall provide stall detection by current sensing on each motor channel.
SUB-E-006	The BMS shall monitor pack voltage, pack current, and pack temperature at a sample rate ≥ 1 Hz.
SUB-E-007	The BMS shall implement a delta-temperature termination algorithm for NiMH fast charging to avoid thermal runaway and overcharge.

ID	Requirement
SUB-E-008	The charging contacts shall be exposed metal pads on the underside of the robot and corresponding pads on the dock; pads shall include alignment chamfers to tolerate ± 10 mm docking misalignment.
SUB-E-009	The charging dock shall include a relay or solid-state switch that energizes the contacts only when the robot is detected as docked.
SUB-E-010	All internal wiring harnesses shall use locking connectors to prevent disconnection due to vibration and shock.

6.5 Firmware and User Interface Subsystem

The firmware and user interface subsystem comprises the embedded firmware framework, the navigation behaviors, the motor control firmware, the BMS firmware, the user input handler, the LED indicator driver, the beeper driver, the fault management logic, the diagnostic logging facility, and the firmware update mechanism.

ID	Requirement
SUB-F-001	The embedded firmware shall provide deterministic task scheduling for sensor sampling and motor control loops.
SUB-F-002	The navigation firmware shall implement at minimum: straight-line cruise, random-bounce on bumper hit, wall-follow with right-side IR sensor, spot-clean spiral, and dock-homing on IR beacon acquisition.
SUB-F-003	The motor control firmware shall ramp drive motor speed from rest to setpoint within 0.5 s without overshoot > 10 %.
SUB-F-004	The BMS firmware shall implement a state-of-charge estimation algorithm accurate to within ± 10 % across battery life.
SUB-F-005	The fault management logic shall transition to a fail-safe state within 100 ms of detected fault.
SUB-F-006	The LED indicator driver shall encode mode, SOC, and fault codes using a documented blink pattern table.
SUB-F-007	The firmware update mechanism shall verify a checksum on the firmware image before applying the update.
SUB-F-008	The diagnostic logging facility shall record ≥ 200 most recent fault events in non-volatile memory.
SUB-F-009	The firmware shall implement schedule retention in non-volatile memory and shall not require user reprogramming after battery replacement.
SUB-F-010	The firmware shall expose a service interface accessible via a wired service port and gated by a service technician authentication mechanism.

7. Test Requirements

Test requirements define the verification activities that establish compliance with the functional, performance, and constraint requirements above. Each test requirement is mapped to one or more requirements in sections 3 through 6 through the Verification Traceability Matrix in section 9.

7.1 Cleaning and Filtration Verification Tests

ID	Test Requirement	Acceptance Criterion	Reference
TR-C-001	Suction power measurement at nominal SOC.	≥ 15 AW.	IEC 60312-1
TR-C-002	Hard floor pickup test using calibrated debris on standardized hard floor surface.	≥ 90 % pickup.	IEC 62885-3
TR-C-003	Low-pile carpet pickup test using calibrated debris on standardized low-pile carpet.	≥ 70 % pickup.	IEC 62885-3
TR-C-004	Edge cleaning reach test with debris placed along a wall.	≥ 25 mm reach beyond chassis.	Internal Test Procedure
TR-C-005	Optional HEPA-class filter efficiency test using monodisperse aerosol challenge.	≥ 99.95 % at MPPS.	EN 1822 / ASTM F1977
TR-C-006	Dust bin retention test under deceleration and reverse motion.	No debris loss back to floor.	Internal Test Procedure
TR-C-007	Filter clog progression test using progressively occluded filter.	Bin-full or filter warning triggers within suction-loss window.	Internal Test Procedure

7.2 Navigation and Mobility Verification Tests

ID	Test Requirement	Acceptance Criterion	Reference
TR-N-001	Coverage test in a standardized 16 m^2 test room over 60 min cleaning.	≥ 80 % coverage measured by overhead optical tracking.	Internal Test Procedure
TR-N-002	Cliff detection test at the edge of a 50 mm drop in multiple approach angles.	Robot stops or reverses before any wheel crosses the edge in 100 of 100 trials.	Internal Cliff Bench
TR-N-003	Bumper response time test using high-speed video and known impact velocity.	Direction change initiated ≤ 200 ms from contact.	High-Speed Camera
TR-N-004	Wall-follow test along a 5 m straight wall.	Robot maintains 30 mm to 50 mm offset over full distance.	Internal Test Procedure
TR-N-005	Threshold climb test using a 15 mm threshold bar.	Robot crosses the threshold in both directions in 95 of 100 trials.	Internal Test Procedure
TR-N-006	Dock-homing test from random start positions within 4 m line-of-sight to dock.	Robot successfully docks within 5 min in 95 of 100 trials.	Internal Test Procedure
TR-N-007	Virtual wall barrier test with the IR barrier emitter active.	Robot does not cross the barrier line in 100 of 100 approaches.	Internal Test Procedure
TR-N-008	Wheel stall detection test using mechanical wheel lock.	Fault declared within 2 s of stall.	Internal Test Procedure
TR-N-009	Brush stall detection test using mechanical brush lock.	Fault declared within 2 s of stall.	Internal Test Procedure
TR-N-010	Lift detection test using rapid lift off floor surface.	All motors stopped within 1 s of lift.	Internal Test Procedure

7.3 Mechanical Verification Tests

ID	Test Requirement	Acceptance Criterion	Reference
TR-M-001	Suction motor life cycle test running motor to specification time plus margin.	≥ 800 h with no functional failure.	Internal Reliability Standard
TR-M-002	Drive motor life cycle test running motor through duty cycle profile to specification time plus margin.	≥ 1,000 h with no functional failure.	Internal Reliability Standard
TR-M-003	Bumper switch life test using automated actuator.	≥ 200,000 actuations without functional failure.	Internal Reliability Standard
TR-M-004	Drop test from 0.5 m onto hardwood, 3 orientations.	No functional failure, no enclosure breach.	IEC 60068-2-31
TR-M-005	Random vibration test on 3 axes for 30 min per axis.	No functional failure, no loose hardware.	IEC 60068-2-64
TR-M-006	Charging contact mate / unmate cycle test using a docking actuator.	≥ 5,000 cycles without exceeding contact resistance specification.	Internal Reliability Standard
TR-M-007	Main brush bristle life test running brush in continuous debris pickup mode.	≥ 200 h before bristle replacement is required.	Internal Reliability Standard
TR-M-008	Side brush bristle life test.	≥ 100 h before bristle replacement is required.	Internal Reliability Standard
TR-M-009	Dust bin latch cycle test.	≥ 5,000 open/close cycles without latch failure.	Internal Reliability Standard
TR-M-010	Caster debris-wrap test using simulated hair and string contamination.	Caster continues to rotate freely after specified contamination exposure.	Internal Test Procedure

7.4 Electrical Verification Tests

ID	Test Requirement	Acceptance Criterion	Reference
TR-E-001	Runtime test on hard floor at full charge until automatic dock entry.	≥ 60 min before dock entry.	Internal Test Procedure
TR-E-002	Runtime test on low-pile carpet at full charge until automatic dock entry.	≥ 45 min before dock entry.	Internal Test Procedure
TR-E-003	Charge time test from 0 % SOC to 100 %.	≤ 3.5 h.	Internal Test Procedure
TR-E-004	Battery cycle life test over 400 full charge / discharge cycles.	Capacity ≥ 80 % of rated at cycle 400.	IEC 61951-2
TR-E-005	Standby current measurement on dock and off dock.	< 10 mA on dock; < 200 µA off dock.	Internal Test Procedure

ID	Test Requirement	Acceptance Criterion	Reference
TR-E-006	NiMH delta-T charging termination verification using thermal probes on battery pack.	Termination occurs at correct delta-T threshold with no overcharge.	Internal Test Procedure
TR-E-007	EMC emissions test on robot and dock in anechoic chamber.	Meets regional emissions limits.	CISPR 14-1
TR-E-008	EMC immunity test on robot and dock.	No functional disturbance at specified field strength.	CISPR 14-2
TR-E-009	ESD immunity test on all user-accessible surfaces.	No latch-up, no data loss, no functional failure.	IEC 61000-4-2
TR-E-010	Charging contact accidental short-circuit test (foreign object across pads when robot off dock).	No spark, no damage; dock contacts de-energized when no robot is detected.	Internal Test Procedure

7.5 Acoustic Verification Tests

ID	Test Requirement	Acceptance Criterion	Reference
TR-A-001	Sound power measurement in Cleaning mode in semi-anechoic chamber.	≤ 68 dB(A).	IEC 60704-2-1
TR-A-002	Beeper sound pressure level measurement at 1 m.	65 to 75 dB(A).	Sound Level Meter
TR-A-003	Sound power measurement on dock in charging state.	≤ 30 dB(A).	IEC 60704-2-1

7.6 Environmental Verification Tests

ID	Test Requirement	Acceptance Criterion	Reference
TR-V-001	Operating temperature test sweeping 5 °C to +40 °C.	Full functional performance across the entire range.	IEC 60068-2-1 and IEC 60068-2-2
TR-V-002	Storage temperature soak from −10 °C to +50 °C.	Full functional performance at room temperature after soak.	IEC 60068-2-1 and IEC 60068-2-2
TR-V-003	Humidity test at 85 % RH for 96 h.	No condensation damage, no functional failure.	IEC 60068-2-78
TR-V-004	Thermal shock test cycling between −10 °C and +50 °C.	No solder joint failure, no functional failure.	IEC 60068-2-14
TR-V-005	Salt fog corrosion test on charging contacts.	No corrosion that impacts contact resistance.	IEC 60068-2-52

7.7 Firmware and User Interface Verification Tests

ID	Test Requirement	Acceptance Criterion	Reference
TR-F-001	Power on to ready-to-clean timing test.	≤ 2 s from button press to motors active.	Internal Test Procedure

ID	Test Requirement	Acceptance Criterion	Reference
TR-F-002	Schedule retention test across battery removal and 30-day storage.	Schedule and last mode retained.	Internal Test Procedure
TR-F-003	Fault injection test for stall, cliff, lift, and battery faults.	Fail-safe state entered within 100 ms.	Internal Test Procedure
TR-F-004	Firmware update reliability test across 500 consecutive updates via service port.	All updates succeed; recovery succeeds in all simulated failures.	Internal Test Procedure
TR-F-005	LED and beeper fault code verification across the documented blink-pattern table.	All codes match table exactly.	Internal Test Procedure
TR-F-006	Long duration stability test running robot at intermittent duty for 7 days.	No firmware hang; watchdog recovery verified for injected fault.	Internal Test Procedure

8. Test Methodology

This section describes the methods by which the test requirements in section 7 are executed. The methodology section answers the question of how each verification activity is performed, what equipment is used, what sample sizes are required, and how data is recorded and analyzed.

8.1 Sample Size Strategy

The sample size for each verification test is determined by the reliability target and the desired confidence level. The general approach is the Weibull demonstration test method. The sample size n required to demonstrate an MTBF of M with confidence C and a Weibull shape parameter β is computed using the standard zero-failure demonstration formula. For example, demonstrating a 1,000 h drive motor life with 90 % confidence and β = 2 requires a sample size of 3 units tested to 2 × the demonstration time without failure.

For statistical navigation tests where pass/fail is binary (e.g., cliff detection, dock homing), the sample size is set by binomial confidence: 100 trials at observed 0 failures supports a one-sided 95 % lower confidence bound on the success rate of approximately 97 %.

8.2 Test Equipment

The test methodology relies on the following classes of equipment. Each equipment item is identified by class rather than by manufacturer or model so that the methodology remains independent of any particular supplier.

Equipment Class	Purpose
Climatic Chamber	Operating and storage temperature, humidity, and thermal shock tests.
Vibration Table	Random and sinusoidal vibration tests on 3 axes.
Drop Tester	Free-fall mechanical shock test from a controlled height.
Anechoic Chamber (Semi-Anechoic)	Sound power and EMC tests.
ESD Simulator	ESD immunity test.
Cliff Test Bench	Controlled-edge step with adjustable height for cliff sensor calibration and verification.

Equipment Class	Purpose
Standardized Test Room	16 m² room with mixed flooring for navigation, coverage, and runtime tests.
Standardized Carpet and Hard Floor Surfaces	Per IEC 62885-3 to support comparable pickup measurements.
Calibrated Debris Sets	Per IEC 62885 — standardized particulate, fiber, and granular debris.
Overhead Optical Tracking System	Coverage and path tracking during navigation tests.
High-Speed Camera	Bumper response time and brush dynamics measurement.
Optical Tachometer	Brush and motor speed measurement.
Power Analyzer	Electrical input power, charging power, and standby current.
Battery Cycler	Charge / discharge cycle life testing.
Aerosol Generator and Particle Counter	HEPA-class filter efficiency tests.
Sound Level Meter and Acoustic Camera	Sound power and source localization tests.
Docking Actuator	Charging contact mate / unmate cycle testing.
Automated Bumper Actuator	Bumper switch cycle life testing.
Thermal Probes	Battery delta-T charging and surface temperature measurement.

8.3 Test Execution Process

Each verification test follows a 5-step execution process. Step 1 is test plan preparation, in which the test owner documents the equipment, sample configuration, environmental conditions, acceptance criteria, and data collection plan. Step 2 is sample preparation, in which the units under test are configured to the specified hardware and firmware revision and assigned serialized identifiers. Step 3 is test execution, in which the test is run according to the plan and all data is captured to the central test data repository. Step 4 is data analysis, in which the recorded data is reviewed against the acceptance criteria and any anomalies are investigated. Step 5 is test report generation, in which the test owner documents the test configuration, results, anomalies, and the pass/fail disposition.

8.4 Test Phases

The verification campaign is organized into 4 sequential phases. Each phase gates entry into the next phase.

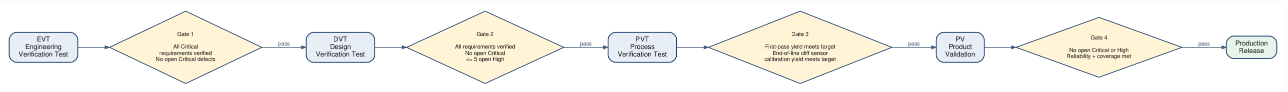

Figure 8-1. Verification Phase Gate Flow.

Phase	Description	Exit Criterion
EVT (Engineering Verification Test)	First functional units; design verification against requirements.	All Critical requirements verified; no open Critical defects.

Phase	Description	Exit Criterion
DVT (Design Verification Test)	Tooled units representative of production; full requirements verification.	All requirements verified; no open Critical defects; ≤ 5 open High defects.
PVT (Process Verification Test)	Pilot production units; manufacturing process verification.	First-pass yield meets target; assembly cycle time meets target; end-of-line cliff sensor calibration yield meets target.
PV (Product Validation)	Production units; final validation including environmental, navigation, life, and field-representative use testing.	No open Critical or High defects; reliability and coverage targets demonstrated.

8.5 Failure Investigation Process

When a verification test produces a failure, the failure investigation process is invoked. The process begins with containment, in which any units exhibiting the failure are quarantined. The process continues with root cause analysis, typically using the 5-Why method, fault tree analysis, or Ishikawa analysis depending on the nature of the failure. For navigation failures specifically, the analysis includes review of the robot's diagnostic log to recover the sequence of sensor events, motor commands, and behavior transitions leading up to the failure. The investigation produces a corrective action plan that may include design change, supplier change, process change, behavior tuning, or test plan change. The corrective action is verified through targeted retest before the verification campaign resumes.

8.6 Data Management

All verification test data is stored in the central test data repository with the following metadata: test ID, test owner, sample serial numbers, hardware revision, firmware revision, test date, equipment used, equipment calibration date, environmental conditions, raw data files, analysis files, disposition, and links to any associated failure investigation records. For navigation tests, the repository additionally stores the overhead-tracking path traces and the robot's diagnostic log for each run. The test data repository is the authoritative record for compliance with the verification requirements.

9. Verification Traceability Matrix

The Verification Traceability Matrix demonstrates that every requirement is addressed by at least one verification test. The matrix is the primary tool used during the verification gate review to confirm that no requirement is left unverified. A representative subset of the matrix is shown below; the full matrix is maintained in the test data repository.

Requirement	Verification Test(s)
FR-001	TR-C-001
FR-003	TR-C-002
FR-004	TR-C-003
FR-101	TR-N-001
FR-102	TR-N-002
FR-103	TR-N-003
FR-104	TR-N-004
FR-106	TR-N-005

Requirement	Verification Test(s)
FR-107	TR-N-006
FR-108	TR-N-007
FR-109	TR-N-008
FR-110	TR-N-009
FR-201	TR-E-001
FR-202	TR-E-002
FR-203	TR-E-003
REQ-R-002	TR-M-001
REQ-R-003	TR-M-002
REQ-R-004	TR-M-003
REQ-S-001	TR-V-001 with surface temperature monitoring
REQ-S-003	TR-N-002
REQ-S-007	TR-N-010
REQ-S-009	TR-N-002, TR-N-008, TR-N-010, TR-V-001, TR-E-007, TR-E-008
REQ-C-001	TR-E-007, TR-E-008
SUB-C-008	TR-C-005
SUB-E-007	TR-E-006
SUB-F-005	TR-F-003
SUB-F-007	TR-F-004

10. Key Functions Summary

The following list of key functions is the high level summary used as a primary input to downstream Failure Mode and Effects Analysis (FMEA) activities. Each key function corresponds to one or more of the functional requirements above and is the level at which failure modes are first enumerated during the analysis.

1. Sweep, agitate, and suction debris into the dust bin across hard floor and low-pile carpet surfaces.
2. Navigate autonomously across an unmapped home environment using random-bounce, wall-follow, and spot-clean behaviors.
3. Detect and avoid floor drop-offs reliably to prevent the robot from falling down stairs or ledges.
4. Detect obstacle contact via the front bumper and recover with a direction change.
5. Move on two driven wheels and a passive caster across thresholds and minor debris without becoming stuck.
6. Operate cordlessly from a rechargeable NiMH battery pack with appropriate safety protections.
7. Return autonomously to the charging dock using an infrared beacon and recharge to full.
8. Respect an infrared virtual wall barrier accessory placed by the user.
9. Allow the user to control modes, start cleaning, and request docking through on-body buttons and an infrared remote.

10. Indicate status, charge state, and fault conditions clearly via LED patterns and audible tones.

11. Open Items and Assumptions

The following items are noted as open or assumed for the purposes of this demonstration document. In a real engineering program, each open item would be tracked to closure through the program decision log.

ID	Item	Type
OPEN-001	Final supplier selection for the NiMH battery pack is to be confirmed.	Open
OPEN-002	Final supplier selection for the drive motors and gearboxes is to be confirmed.	Open
OPEN-003	Final design of the dock IR beacon waveform and the virtual wall waveform is to be confirmed; the two emitters must not interfere with each other or with the remote control.	Open
OPEN-004	The exact threshold for the dirt detection sensor that triggers a spot-clean spiral is to be tuned during DVT.	Open
ASM-001	Target sales regions are assumed to be North America, Europe, and East Asia.	Assumption
ASM-002	The robot is assumed to be used in clean, dry indoor environments only; outdoor use, wet floors, and wet debris pickup are explicitly out of scope.	Assumption
ASM-003	The home environments encountered are assumed to have predominantly hard floors and low-pile carpet; deep-pile carpet performance is not a program target.	Assumption
ASM-004	The annual production volume is assumed to be approximately 750,000 units.	Assumption
ASM-005	The robot is not intended to collect hazardous, flammable, or hot debris; warning labels and user manual will reflect this.	Assumption

12. References

1. IEC 60312-1 — Vacuum cleaners for household use — Methods of measuring performance. https://webstore.iec.ch/publication/22013
2. IEC 62885 series — Surface cleaning appliances. https://webstore.iec.ch/publication/26152
3. IEC 60335-2-2 — Household appliance safety — Vacuum cleaners. https://webstore.iec.ch/publication/63645
4. IEC 60335-2-108 — Household appliance safety — Robotic cleaners (battery-operated). https://webstore.iec.ch/publication/65789
5. IEC 60704-2-1 — Acoustic noise test code for vacuum cleaners. https://webstore.iec.ch/publication/26352
6. IEC 60068 series — Environmental Testing. https://webstore.iec.ch/publication/505
7. IEC 60529 — Degrees of Protection Provided by Enclosures. https://webstore.iec.ch/publication/2452
8. IEC 62133-2 — Safety requirements for portable Li-Ion cells. https://webstore.iec.ch/publication/32662
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10. UN 38.3 — Transport tests for lithium batteries. https://unece.org/transport/dangerous-goods
11. ANSI/AHAM VC-1 — Method of measuring vacuum cleaner performance. https://www.aham.org

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14. ISO 5725 — Accuracy of measurement methods and results. https://www.iso.org/standard/69419.html
15. Engineering Program Management Tool 1 — AI-Powered FMEA Assistant. https://epm-tool-fmea.vercel.app

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End of Engineering Requirements Specification ERS-GEN-RVC-100, Revision A.